

DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

Enhancing TA-LLM dialogue flow — slot-filling and tool rejection —
via automatic paired trajectory construction from a 5-state MDP.

RLHF / DPO

Tool-Augmented LLMs

NLP • ML Research

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

The three failure modes of SFT-only training for TA-LLMs

1

Missing Follow-Up Questions

SFT models skip asking for missing arguments and proceed with incomplete information.

2

Hallucinated Tool Invocations

Without slot-filling, models fabricate parameter values — silently producing wrong tool calls.

3

Failure to Reject Out-of-Scope Calls

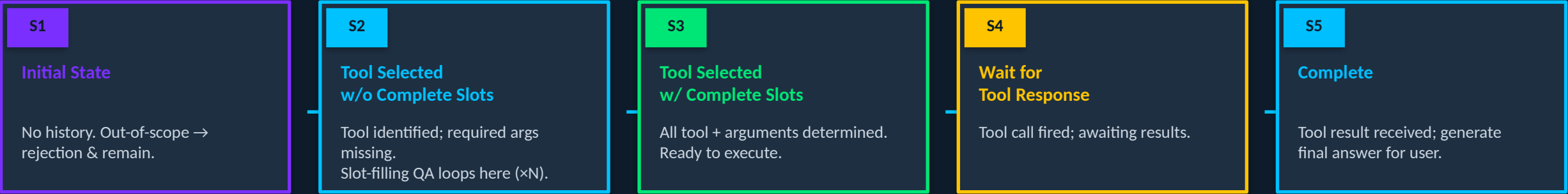
SFT models attempt tool calls even when no suitable tool exists, instead of returning a rejection.

Root cause: SFT only mimics expert trajectories — it never sees or learns from incorrect dialogue flows. No penalty for hallucinated or missing tool interactions.

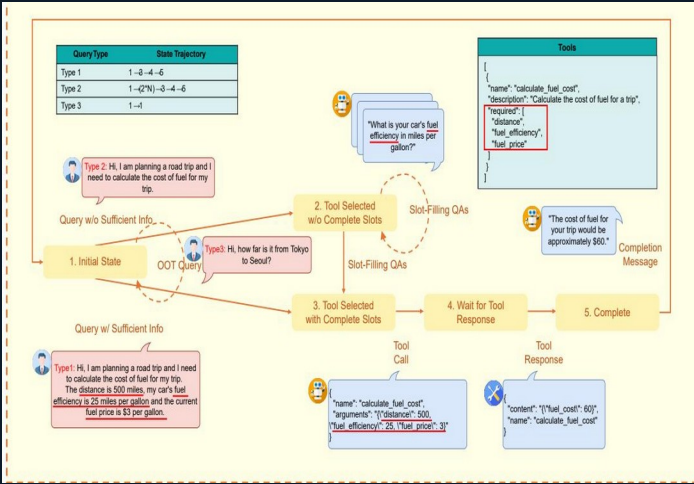
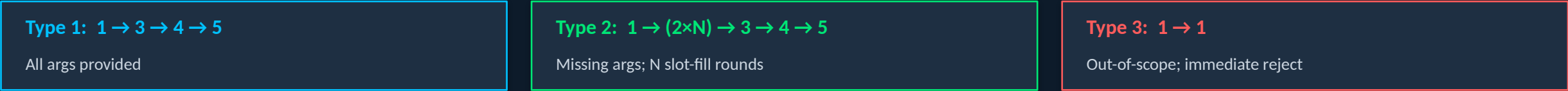
→ **Solution: Direct Preference Optimization (DPO) with paired correct/incorrect trajectories**

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

Each conversation turn is a state transition — reward comes from reaching State 5



State Trajectories by Query Type:



↑ Figure 1 — Five internal states and state trajectories (source: arxiv 2504.02882)

Three Query Types Define Distinct State Trajectories

Query classification drives which preference pairs are constructed

Query Type	State Trajectory	Behaviour	Example
Type 1	1 → 3 → 4 → 5	All required arguments present; no slot-filling needed.	"Book a flight from Seoul to Tokyo on Dec 5th" → all slots filled → immediate call
Type 2	1 → (2×N) → 3 → 4 → 5	One or more args missing; N rounds of slot-filling QA.	"Book a flight to Tokyo" → missing: origin, date → 2 follow-up Qs → call
Key insight: Each query type requires a different 'preferred' trajectory. Misclassification (e.g., skipping slot-filling for Type 2 or calling a tool for Type 3) generates the 'rejected' trajectory — the negative example in DPO.			

Automatic Paired Trajectory Construction Without Human Labels

MDP structure enables fully automated chosen / rejected pair generation

Query	Chosen Trajectory	Rejected Trajectory	Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2*M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562	Easy/Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2*N)→3→4→5	1→(2*M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089 / 562	Easy/Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2*N)→3→4	Tool call reject	562	Hard

No human annotation required — incorrect trajectories are derived algorithmically from the MDP structure.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

FunctionChat-Bench evaluation — metrics: Call, Completion, Slot, Relevance, Micro/Macro Avg

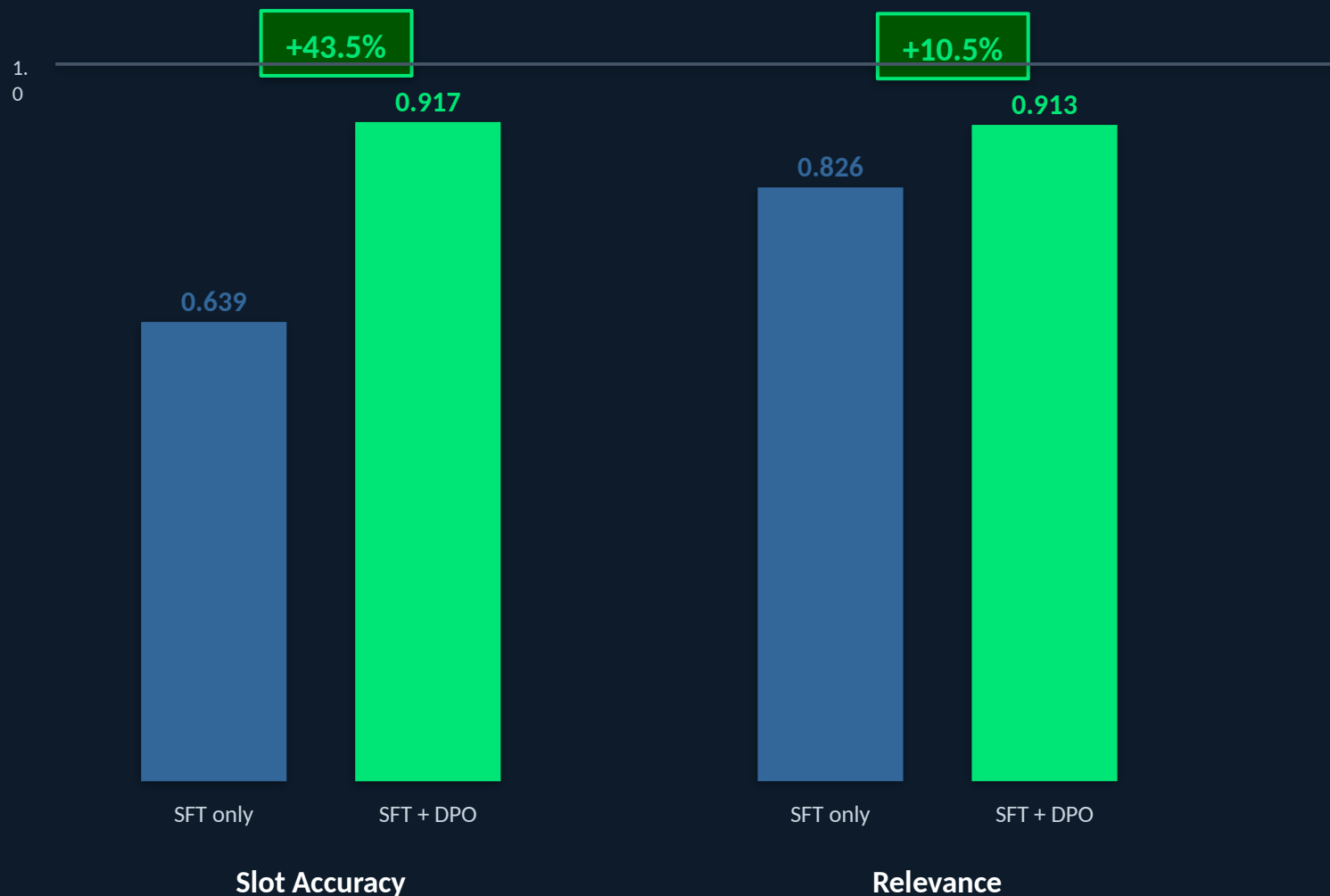
Model	Method	Call	Compl.	Slot	Relev.	Micro	Macro
Prop.-8B	DiaTool-DPO only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

= SFT + DiaTool-DPO (best Macro Avg row per model)

SFT + DiaTool-DPO wins Macro Avg for all three model families.

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Adding DPO to SFT unlocks dramatic dialogue-control gains on an open-source model



Why such large gains?

- ▶ Slot accuracy: SFT cannot teach when NOT to invent slot values.
- ▶ DPO directly penalises hallucinated slot trajectories (Type 2 pairs).
- ▶ Relevance (rejection): SFT models rarely see out-of-scope examples.
- ▶ DPO Type 3 pairs teach refusal with explicit negative signal.
- ▶ Macro Avg: 0.816 → 0.904 (+10.8pp overall)

Near GPT-4o level: 94.8% info-gathering • 91% tool rejection

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

DiaTool-DPO — findings and implications for deployment

1

MDP formulation enables systematic trajectory pairing

5 internal states + 3 query types → automatic chosen/rejected pairs without any human labeling.

2

DPO complements SFT — never replaces it

DiaTool-DPO-only performs poorly. SFT provides the base capability; DPO sharpens dialogue control.

3

Slot-filling and tool rejection see largest gains

LLaMA-3-8B: Slot +43.5% (0.639 → 0.917) and Relevance +10.5% (0.826 → 0.913) vs SFT-only.

4

Language-agnostic method — broad production applicability

Primary experiments in Korean; validated in English — the MDP+DPO framework is domain-general.

5

Opens path to GPT-4o-level performance on smaller open models

Achieves ~94.8% information-gathering and ~91% tool-rejection accuracy — competitive with GPT-4o.